

Philipp Gadow

Publication list

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- Bibliometric information: <https://inspirehep.net/author/profile/P.P.Gadow.1>
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As a member of the CMS Collaboration and former member of the ATLAS Collaboration, I am a co-author of over 675 peer-reviewed journal articles. I included those where I have made a substantial contribution below. The full list is available through <https://inspirehep.net>, which also serves as the source for the number of citations of the publications (updated on 29.06.2026).

Selection of five most significant publications

- 2026 Transforming jet flavour tagging at ATLAS
Nature Commun. 17 (2026) 1, 541, 92 citations (29.06.2026)
Personal contributions: co-coordination of the project (with 2 others), editor of introduction and performance sections, performance studies.
- 2024 Search for heavy resonances in four-top-quark final states in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector
Eur. Phys. J. C 84 (2024) 157, arXiv:2304.01678, 26 citations (29.06.2026)
Personal contributions: leadership of analysis, statistical analysis, signal simulation, analysis software.
- 2023 ATLAS flavour-tagging algorithms for the LHC Run 2 pp collision dataset
Eur.Phys.J.C 83 (2023) 7, 681, arXiv:2211.16345, 445 citations (29.06.2026)
Personal contributions: leadership of the publication, performance studies.
- 2021 Search for dark matter produced in association with a Standard Model Higgs boson decaying into b -quarks using the full Run 2 dataset from the ATLAS detector
JHEP 11 (2021) 209, arXiv:2108.13391, 92 citations (29.06.2026)
Personal contributions: analysis software maintenance, derivation software and requests, optimisation of missing transverse momentum significance, study of tight-jet cleaning.
- 2021 Search for dark matter produced in association with a dark Higgs boson decaying to $W^\pm W^\mp$ or ZZ in fully hadronic final states using pp collisions at $\sqrt{s} = 13$ TeV recorded with the ATLAS detector
Phys. Rev. Lett. 126, 121802, arXiv:2010.06548, 43 citations (29.06.2026)
Personal contributions: main analyser (of 2), analysis software development and maintenance, analysis strategy, track-assisted-reclustered jet optimisation, statistical analysis, signal simulation.

Published peer-reviewed journal articles

- 2026 H. Bahl, P. Gadow, R. Kumar, K. Peters, P. Stylianou, G. Weiglein. Complementarity of di-top and four-top searches in interpreting possible signals of new physics
arXiv:2602.15027 (submitted to *JHEP*), 1 citation (29.06.2026)
- 2026 R. Ballabriga, P. Gadow et al. Characterization of the H2M monolithic CMOS sensor
Nucl.Instrum.Meth.A 1086 (2026) 171353, 1 citation (29.06.2026)

- 2025 J. Barr, D. Biswas, M. Draguet, P. Gadow et al. Salt: Multimodal Multitask Machine Learning for High Energy Physics
J. Open Source Softw. 10 (2025) 112, 7217, 3 citations (29.06.2026)
- 2024 F. Riti, P. Gadow, D. Walter, M. Vieites Diaz, B. M. Latacz et al. A history of CERN in seven physics milestones
Nature Rev. Phys. 6 (2024) 10, 582-586, 0 citations (29.06.2026)
- 2024 S. Barr, P. Gadow et al. Umami: A Python toolkit for jet flavour tagging
Journal of Open Source Software, 9(102), 5833, 2 citations (29.06.2026)
- 2024 Search for top-philic heavy resonances in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector
Eur. Phys. J. C 84 (2024) 157, arXiv:2304.01678, 26 citations (29.06.2026)
- 2023 Search for single vector-like B quark production and decay via $B \rightarrow bH(b\bar{b})$ in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector
JHEP 11 (2023) 168, arXiv:2308.02595, 31 citations (29.06.2026)
- 2023 ATLAS flavour-tagging algorithms for the LHC Run 2 pp collision dataset
Eur. Phys. J. C 83 (2023) 7, 681, arXiv:2211.16345, 445 citations (29.06.2026)
- 2021 Search for dark matter produced in association with a Standard Model Higgs boson decaying into b -quarks using the full Run 2 dataset from the ATLAS detector
JHEP 11 (2021) 209, arXiv:2108.13391, 92 citations (29.06.2026)
- 2021 Search for dark matter produced in association with a dark Higgs boson decaying to $W^\pm W^\mp$ or ZZ in fully hadronic final states using pp collisions at $\sqrt{s} = 13$ TeV recorded with the ATLAS detector
Phys. Rev. Lett. 126, 121802, arXiv:2010.06548, 43 citations (29.06.2026)
- 2019 Combination of Searches for Invisible Higgs Boson Decays with the ATLAS Experiment
Phys. Rev. Lett. 122, 231801, arXiv:1904.05105, 219 citations (29.06.2026)
- 2019 Constraints on mediator-based dark matter and scalar dark energy models using $\sqrt{s} = 13$ TeV pp collision data collected by the ATLAS detector
JHEP 05 (2019) 142, arXiv:1903.01400, 190 citations (29.06.2026)
- 2018 Search for dark matter in events with a hadronically decaying vector boson and missing transverse momentum in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector
JHEP 10 (2018) 180, arXiv:1807.11471, 156 citations (29.06.2026)

Conference Publications

The ATLAS Collaboration also produces internally-reviewed "conference notes" and "public notes" in advance of conferences, so that results may be discussed with colleagues in other experiments and in the theory community. Conference notes superseded by a peer-reviewed publication are not included.

- 2024 Z. Bhatti, K. Cranmer, I. Espejo, L. Heinrich, P. Gadow, P. Rieck, J. von Ahnen. Efficient Search for New Physics Using Active Learning in the ATLAS Experiment
EPJ Web Conf. 295 (2024) 09013
- 2022 Active Learning reinterpretation of an ATLAS Dark Matter search constraining a model of a dark Higgs boson decaying to two b -quarks
ATL-PHYS-PUB-2022-045 (non-peer-reviewed ATLAS publication)
- 2021 S. Malik, P. Gadow et al. Software Training in HEP
Comput. Softw. Big Sci. 5 (2021) 1, 22, arXiv:2103.00659
- 2019 D. Cieri, P. Gadow et al. A Lightweight First-Level Muon Track Trigger for Future Hadron Collider Experiments
PoS TWEPP2018 (2019) 051

- 2019 S. Abovyan, P. Gadow et al. First-level muon track trigger for future hadron collider experiments
Nucl.Instrum.Meth.A 936 (2019) 321-322
- 2019 RECAST framework reinterpretation of an ATLAS Dark Matter Search constraining a model of a dark Higgs boson decaying to two b -quarks
ATL-PHYS-PUB-2019-032 (non-peer-reviewed ATLAS publication)
- 2018 Search for Dark Matter Produced in Association with a Higgs Boson Decaying to $b\bar{b}$ at $\sqrt{s} = 13$ TeV with the ATLAS Detector using 79.8 fb^{-1} of pp collisions
ATLAS-CONF-2018-039 (non-peer-reviewed ATLAS publication)
- 2018 P. Gadow. Search for dark matter produced in association with a hadronically decaying Z' vector boson with the ATLAS detector at the LHC
PoS LHCP2018 (2018) 016
- 2017 S. Abovyan, P. Gadow et al. Hardware Implementation of a Fast Algorithm for the Reconstruction of Muon Tracks in ATLAS Muon Drift-Tube Chambers for the First-Level Muon Trigger at the HL-LHC
Proceedings, 2017 IEEE NSS/MIC
- 2016 P. Gadow, O. Kortner, S. Kortner, H. Kroha, F. Müller, R. Richter. Performance of a First-Level Muon Trigger with High Momentum Resolution Based on the ATLAS MDT Chambers for HL-LHC
Proceedings, 2015 IEEE NSS/MIC, arXiv:1511.09210
- 2016 S. Nowak, P. Gadow et al. Optimisation of the Read-out Electronics of Muon Drift-Tube Chambers for Very High Background Rates at HL-LHC and Future Colliders
Proceedings, 2015 IEEE NSS/MIC

Technical Design Reports

- 2017 ATLAS Collaboration. Technical Design Report for the Phase-II Upgrade of the ATLAS Muon Spectrometer
CERN-LHCC-2017-017